

Outset · AI in Healthcare · Day 3 · Group 3

Stroke on a brain CT -- and the fairness the data hides

A working stroke detector you can't audit, and a tabular model you can audit and fix -- because it records who's in it.

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Outset

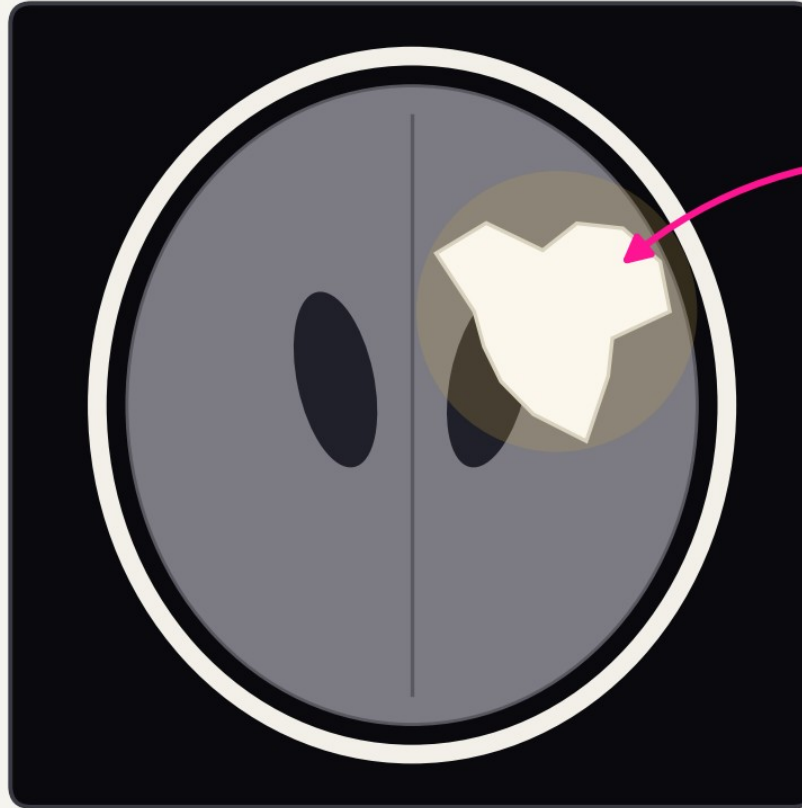
2026-07-08

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BACKGROUND

A stroke is a race against the clock

A stroke on a head CT is a race against the clock



acute hemorrhage

bright = hyperdense blood

TIME IS BRAIN



*~1.9 million neurons
lost every minute*

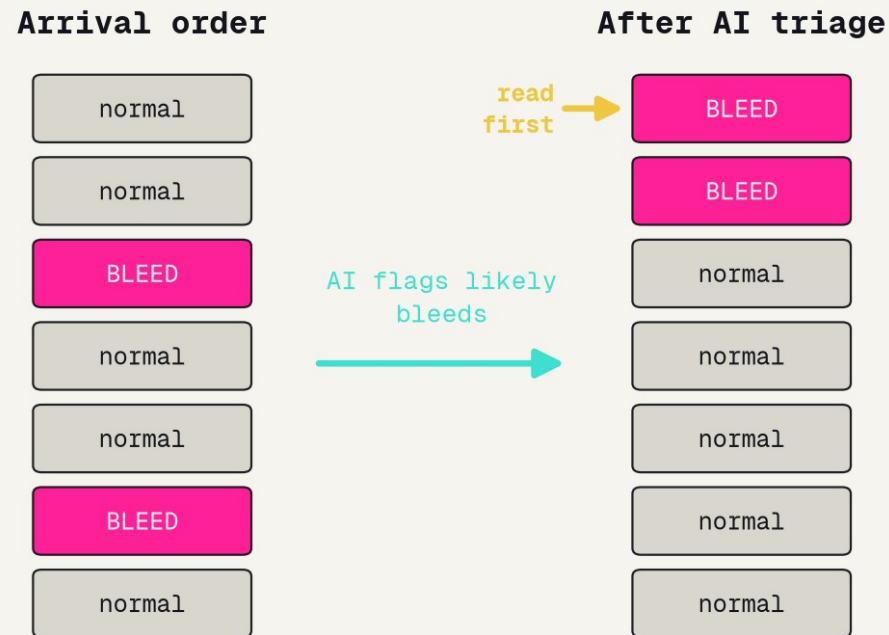
so the bleed has to be
found -- and read -- first

hemorrhage appearance per Chilamkurthy 2018 -- 'time is brain' per Saver 2006

A fresh bleed is bright (hyperdense) on CT -- and every minute of delay costs about 1.9 million neurons, so it must be found and read first.

Triage AI reads the dangerous scans first

SORT, DON'T JUST QUEUE



A radiologist faces a long list of scans. Read in arrival order, a bleed can wait behind ten normal heads.

Triage AI scores every scan and lifts the likely bleeds to the TOP, so the dangerous ones are read first.

Real tools do this with AUCs above 0.90 and cut time-to-treatment.

Real triage tools move likely bleeds to the top of the list, cutting time-to-treatment.

adapted from Chilamkurthy et al. 2018 / Seyam et al. 2022

The catch: a dataset can hide who is in it

A model can work on average and still hide who it fails.

WORKS ON AVERAGE

A detector can hit a strong overall AUC and still be much worse for one group of patients -- a hidden failure.

BUT UN-CHECKABLE

If age, sex, race, and scanner were never recorded, there is nothing to group by. The audit that would catch it cannot even run.

THE DATA

Two datasets, documented very differently

BRAIN CT

normal vs. stroke

Real head-CT slices, one label each. That is ALL the dataset records -- no age, sex, race, or scanner.

you can train it -- but not audit who it fails

TABULAR STROKE

5,109 patients

One row per patient, and it records sex, age, hypertension, glucose, and smoking -- the facts an audit needs.

you can audit AND fix it

What a datasheet should record

SIX QUESTIONS

A datasheet asks who is in the data -- age, sex, race, scanner, label source, consent -- so you know its limits before you trust it.

TWO ANSWERS

Our brain-CT set answers only two: the image, and a stroke / normal label. The four demographic rows are simply blank.

What a datasheet asks for

- ☒ Age
- ☒ Sex
- ☒ Race / ethnicity
- ☒ Scanner / site
- ☒ Label source
- ☒ Patient consent

What our brain-CT dataset has

- ☒ CT image
- ☒ Stroke / normal label
- ☐ Age
- ☐ Sex
- ☐ Race / ethnicity
- ☐ Scanner / site

Four of the six questions have no answer -- so we cannot check who the model fails.

concept from Gebru et al. 2021 (Datasheets for Datasets) / Tripathi et al. 2023

THE MODEL

Borrow a brain: transfer learning for the scans

PRETRAINED

a CAFormer already trained on millions of everyday photos

it knows edges, texture, shape



FINE-TUNE

freeze that body; train only a small new head on our 224px CT slices

it learns the look of a bleed



DETECT

ask the fine-tuned network for one call on a new scan

normal, or stroke

Borrowed knowledge is why a network can learn stroke from a few hundred scans, not a few million.

The CT recipe, so it is reproducible

1

MODEL

caformer_s18, pretrained, backbone FROZEN, a fresh 2-class head trained on top

2

RESIZE

each CT slice decoded to 3-channel color and resized to 224 x 224 pixels

3

NORMALIZE

pixels rescaled to the mean/scale the backbone was trained on; light flips augment TRAIN only

4

BALANCE

stroke vs. normal framed as one binary call, graded on sensitivity, not just accuracy

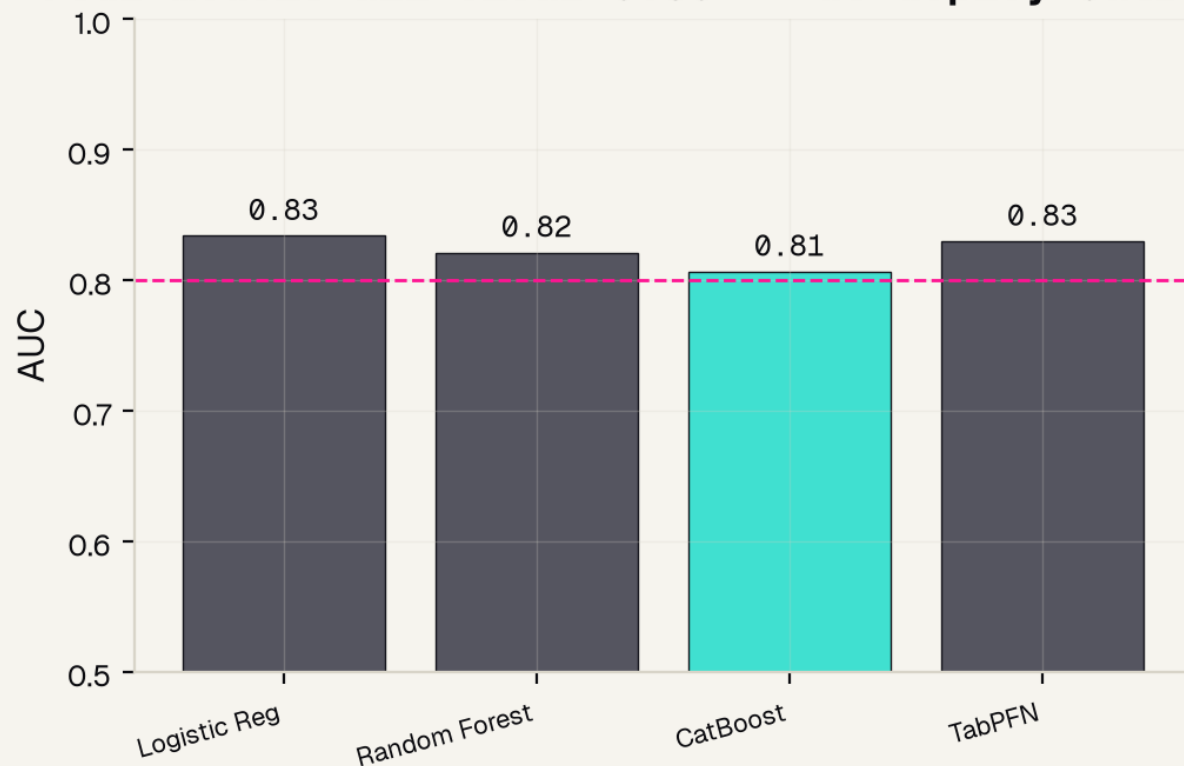
5

SPLIT

separate train / val / test; every number we report is on the held-out test set

For the table: we ran a bake-off

Four models all clear 0.80 -- we deploy CatBoost



WE TESTED, WE DID NOT GUESS

0.81

CatBoost AUC. Four models all cleared 0.80, so the honest choice is not the top decimal.

We deploy CatBoost because it is a tree we can explain with SHAP and audit by sex.

RESULTS

The CT detector works

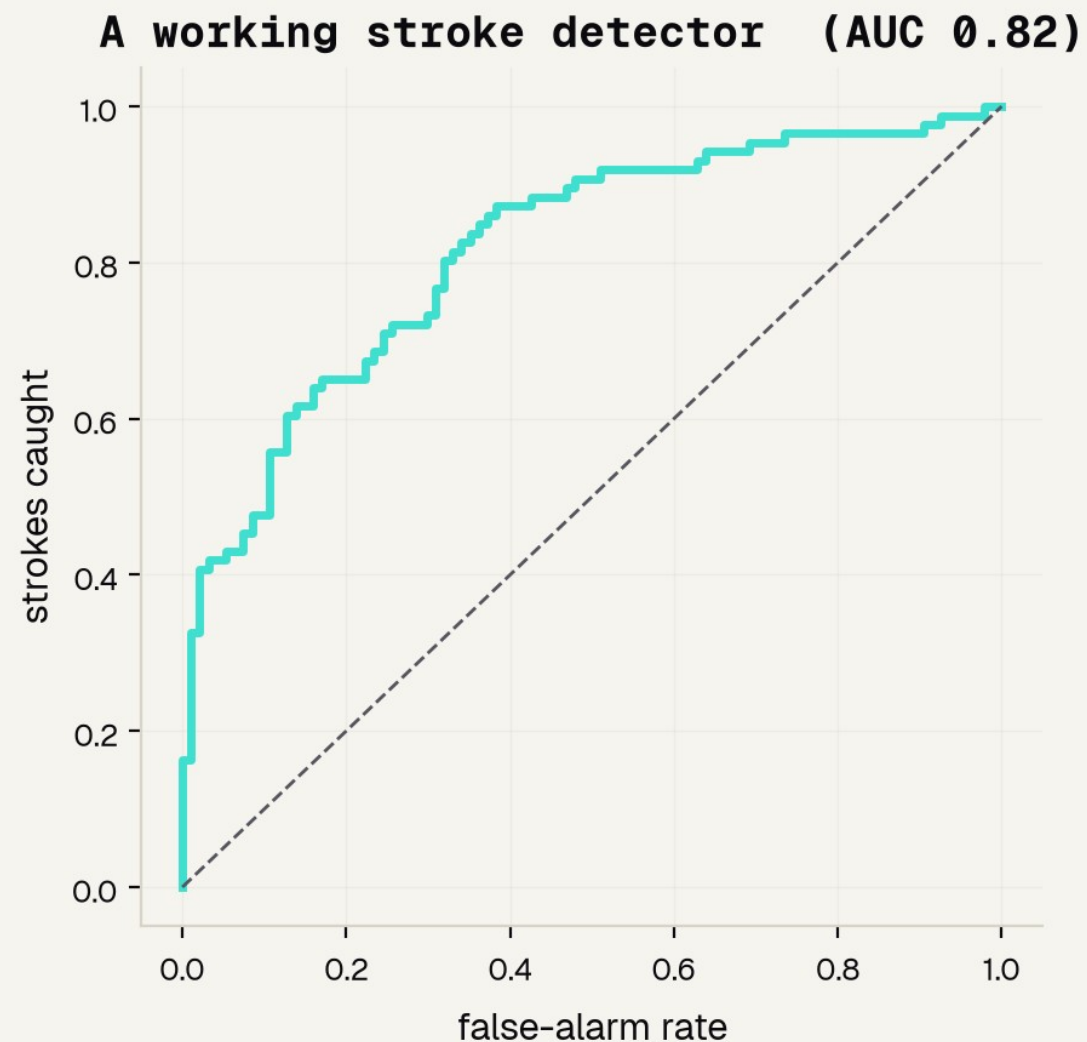
0.817

AUC -- the area under the ROC curve

The ROC curve sweeps every cut-off and plots strokes caught against false alarms. AUC is the area underneath.

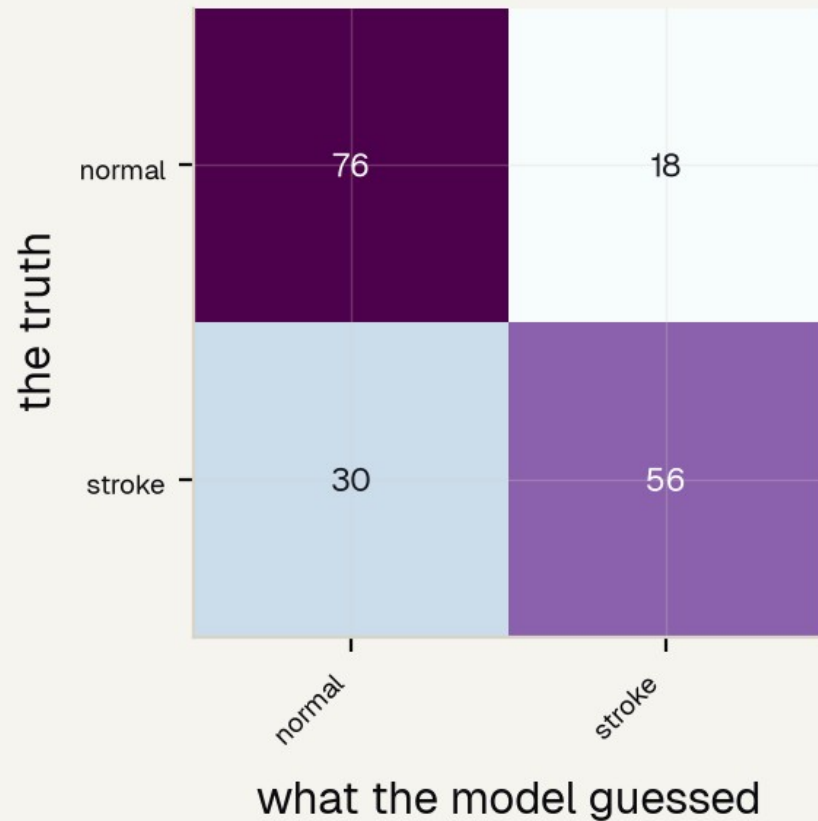
0.5 = a coin flip 1.0 = perfect

The curve hugs the top-left corner -- a genuinely working detector.



Read the two errors apart

Stroke detector: test accuracy 0.73



65%

of real STROKES caught
(sensitivity) -- most, but not all

81%

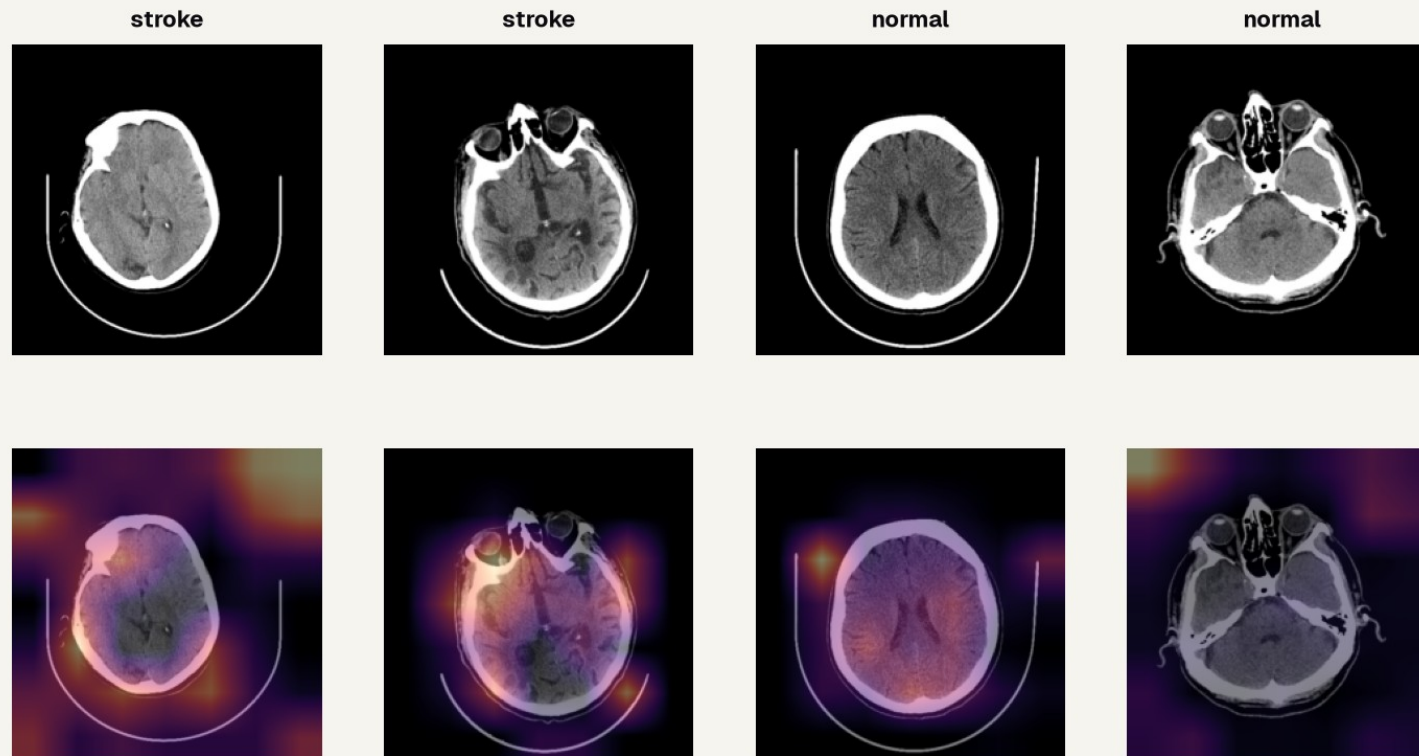
of real NORMALS cleared
(specificity) -- few false alarms

the
dial

for a safety-net triage tool, lower
the threshold to catch more
strokes, accepting more false
alarms

Is it reading the brain, or cheating off the skull?

Is it reading the brain, or cheating off the edges?



Grad-CAM = feature importance for an image: some heat sits on brain tissue (good), but some drifts to the SKULL EDGE and border -- a shortcut that works here and would break on a new scanner.

The wall: the audit we cannot run

We can audit by CLASS (normal vs. stroke) because the label exists. But the audit that matters -- older vs. younger, women vs. men, one scanner vs. another -- we cannot run at all.

Not because it is hard: the dataset never recorded those fields, so there is nothing to group by.

What this dataset tells us about each patient

GIVEN: the CT image + normal / stroke label

MISSING:

✗ age	✗ sex
✗ race / ethnicity	✗ scanner model
✗ hospital / site	✗ time since injury

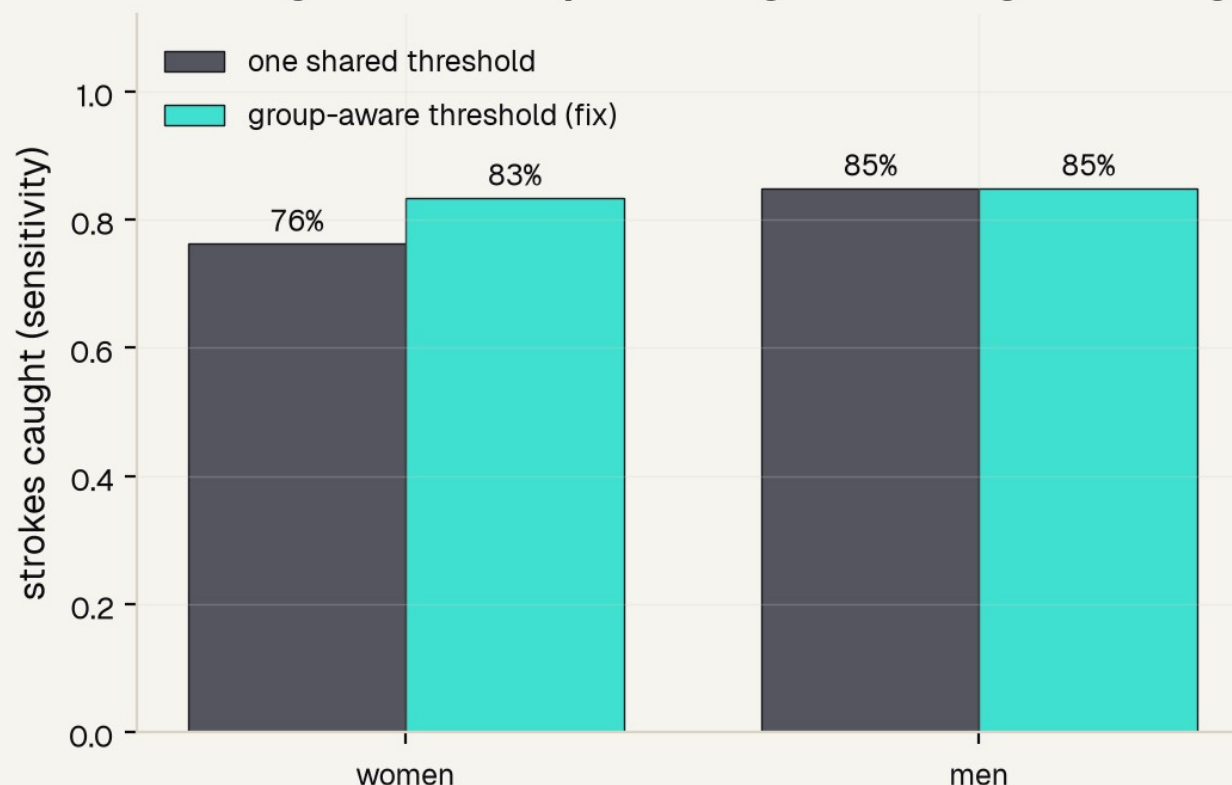
You cannot audit fairness across groups the data never records.

DISQUALIFYING, NOT A FOOTNOTE

The detector might be far worse for some group of patients, and we would have no way to know. For a tool that touches patients, an un-runnable fairness audit sinks it on its own.

Now a dataset that records sex: audit, then fix

Auditing -- and equalizing -- who gets caught



76% → 83%

women's strokes caught, after a group-aware threshold (men held at 85%)

gap 9% → 2%

recording sex made the gap visible AND closable

The audit the CT scans made impossible -- run, and fixed.

What drives the risk score: age dominates

SHAP = feature importance for a table: how much each column moves the prediction.

age

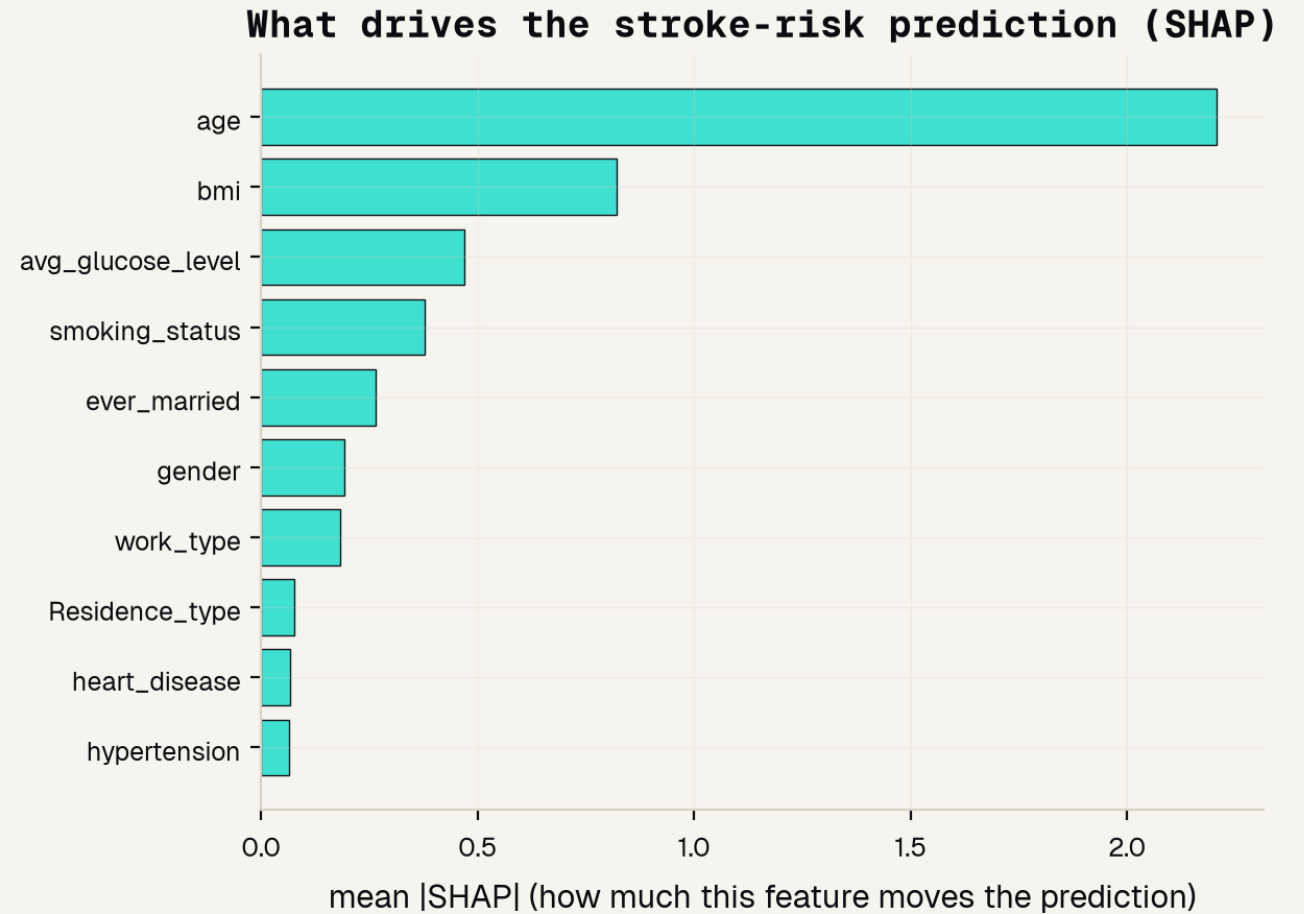
by far the strongest driver

bmi + glucose

the next tier of risk

sex

barely moves the score -- yet it mattered for FAIRNESS



THE TAKEAWAY

Recording demographics makes fairness auditable - and fixable

CT: WORKS, CAN'T AUDIT

AUC 0.817 -- a real detector. But the dataset records only image + label, so the who-does-it-fail audit simply cannot run.

no demographics = no audit

TABULAR: AUDITED + FIXED

AUC 0.81, and because it records sex we found a real gap (76% vs 85%) and closed it (to 83% vs 85%) with a group-aware threshold.

metadata = auditable +
fixable

THE RULE

A model that runs is easy. A model you can AUDIT is the real bar -- and recording who is in the data is what makes it possible.

audit is the bar

References

HEAD-CT TRIAGE AI

- [1] Chilamkurthy 2018, The Lancet -- critical findings on head CT.
- [2] Flanders 2020, Radiology: AI -- the RSNA hemorrhage challenge.
- [3] Seyam 2022, Radiology: AI -- a deployed tool in the clinic.

DATA & DOCUMENTATION

- [4] Gebru 2021, Comms of the ACM -- Datasheets for Datasets.
- [5] Tripathi 2023, J Am Coll Radiol -- biases in radiology-AI data.
- [8] Kim 2022, BMC Med Imaging -- transfer learning.

FAIRNESS NEEDS METADATA

- [6] Seyyed-Kalantari 2021, Nature Medicine -- underdiagnosis bias.
- [7] Yang 2024, Nature Medicine -- limits of fair imaging AI.
- Saver 2006, Stroke -- "Time Is Brain".

The honest bottom line

An imaging dataset can hide who is in it.



Recording demographics is what turns fairness from a hope into something you can measure -- and fix. That is the finding, not the footnote.

Thank You

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夢想